

SENIOR SOFTWARE ENGINEER

AI Systems · Backend Architecture · Production Infrastructure

Document: Technical Engineering Portfolio
Focus: AI Systems & Backend Architecture
Audience: CTOs, Engineering Managers & Founders
Specialization: Vibe Code → Production Ready

Executive Summary: Senior Software Engineer specializing in taking rapidly built or vibe-coded applications and re-engineering them into clean, reliable, production-ready systems. Deep expertise across FastAPI backend microservices, high-throughput vLLM model serving, adaptive RAG pipelines, enterprise Microsoft Entra ID / OAuth 2.0 security, and automated CI/CD.

1. ENGINEERING CASE STUDIES (FLAGSHIP WORK)

CASE STUDY 01: Enterprise LLM Application [Confidential Client Work]
Domain: Australian Engineering Consultancy (Cerecon) | **Key Outcome:** Turnaround reduced from Days → < 1 Hour

The Problem: Automated proposal generation needed to convert raw client briefs into audit-ready fee proposals with automated rate cards and branded Word/Excel deliverables without risking hallucinated pricing or audit gaps.

Engineering Implementation:

- **Backend Architecture:** Built and maintained FastAPI services on Azure App Service with strict domain separation.
- **Multi-Stage Prompt Assembly:** Engineered structured context preservation across multi-phase LLM runs.
- **Security & Identity:** Implemented per-user OAuth 2.0 token handshakes with Total Synergy v4 API and Microsoft Entra ID RBAC across 24 mapped enterprise use cases, eliminating shared-account audit gaps.
- **DevOps:** Solved Azure App Service limits using Kudu async zipdeploy with zero commit loss migration.

Tech Stack: Python, FastAPI, Azure App Service, Azure DevOps, Microsoft Entra ID, OAuth 2.0, RBAC, REST APIs, LLMs

CASE STUDY 02: High-Throughput LLM Inference Service [AI Infrastructure]
Hardware Target: Single NVIDIA RTX 2080 Ti | **Key Outcome:** 100 Concurrent Requests Supported

The Problem: Deploying production LLM serving on constrained GPU hardware while handling high concurrent user loads without VRAM out-of-memory crashes or high time-to-first-token (TTFT) queue spikes.

Engineering Implementation:

- **vLLM Engine Integration:** Leveraged continuous batching and PagedAttention to eliminate memory fragmentation.
- **KV-Cache Optimization:** Tuned memory allocation chunks and asynchronous request queue buffers.
- **Pipeline Flow:** Client → API Gateway → Async Queue / Scheduler → vLLM Batcher → GPU VRAM → Streamed Tokens.
- **Benchmarking:** Profiled token generation rates under synthetic load to maintain sub-second first-token response.

Tech Stack: Python, vLLM, Continuous Batching, PagedAttention, PyTorch, FastAPI, Docker, GPU Profiling

CASE STUDY 03: Contextual Dynamic RAG Framework [Search & LLM Infrastructure]
Architecture: Hybrid Dynamic Routing | **Key Outcome:** +60% Retrieval Efficiency Improvement

The Problem: Standard naive RAG pipelines suffer from high latency on exact-match queries and poor context quality on complex multi-hop reasoning over heterogeneous enterprise documents.

Engineering Implementation:

- **Dynamic Query Router:** Classifies incoming queries by intent, keyword density, and semantic complexity.
- **Hybrid Retrieval Layer:** Routes between FAISS dense vector search and BM25 sparse keyword indices dynamically.
- **Cross-Encoder Reranking:** Applied fine-grained reranking over candidate chunks before prompt context assembly.
- **Compaction:** Deduplicated context windows to maximize answer precision and minimize LLM inference tokens.

Tech Stack: FAISS, BM25, Cross-Encoder, LangGraph, Python, FastAPI, Hugging Face Embeddings

2. TECHNICAL STACK & CORE COMPETENCIES

Languages	Python, C++, SQL, JavaScript, HTML/CSS, Bash
AI / LLM Systems	LoRA / PEFT, Unsloth, vLLM, llama.cpp, GGUF, Dynamic RAG, Prompt Engineering, LangChain, LangGraph, Hugging Face Transformers, OpenAI / Anthropic / Gemini APIs, Coqui TTS
ML / Computer Vision	PyTorch, TensorFlow, TensorFlow Lite, Ultralytics YOLO (v8), OpenCV, scikit-learn, Optuna, Weights & Biases, Pandas, NumPy
Backend & Cloud	FastAPI, REST APIs, Docker, Git, Azure App Service, Azure DevOps, Microsoft Entra ID, OAuth 2.0, RBAC, CI/CD Pipelines, Hugging Face Spaces
Databases & Analytics	PostgreSQL, MySQL, MongoDB, Supabase, FAISS Vector Index, Power BI, Tableau

3. PUBLIC PROJECTS & OPEN-SOURCE ARTIFACTS

Kisan Rabta Voice-to-Insight Agricultural AI Platform	Fine-tuned OpenAI Whisper on 18,000 scraped text samples & 2 hours of regional Urdu audio. Extracted structured agricultural information & crop disease diagnostics via custom NER pipelines, exposed via low-latency REST APIs.	Stack: Whisper, NER, Python, FastAPI, React Native
Medical Reasoning LLM DeepSeek R1 Distill LLaMA 8B	Fine-tuned DeepSeek R1 Distill LLaMA 8B on clinical diagnostic reasoning datasets using Unsloth 4-bit QLoRA to maximize VRAM efficiency. Published model weights for public research and experimentation.	Stack: DeepSeek, LLaMA, LoRA, Unsloth, PyTorch
ID Document Parsing Structured National Identity Extraction	Computer vision and OCR pipeline for localized bounding box detection (Name, ID number, DOB, Barcode/QR) with automated cross-field checksum validation deployed to interactive web spaces.	Stack: YOLOv8, OCR, OpenCV, Python, HF Spaces
Automatic Number Plate Recognition (ANPR) Real-Time Vehicle Localization	High-accuracy object detection achieving 97.5% Precision and 97.5% Recall on benchmark splits across challenging lighting and video stream angles.	Stack: YOLOv8, OpenCV, Python, HF Spaces

4. PROFESSIONAL EXPERIENCE & EDUCATION

Senior Software Engineer Australian Engineering Consultancy Jun 2026 — Present	Lead engineer on AI enterprise software, backend architecture, third-party integrations, and Azure deployments. Rebuilt prompt-assembly layer to reduce proposal turnaround from days to <1 hr; implemented Microsoft Entra ID RBAC across 24 use cases and per-user OAuth 2.0 with Total Synergy v4 API.
Software Engineer — AI Applied AI Consultancy Jun 2025 — Jun 2026	Shipped generative AI, vLLM inference services (100 concurrent requests on RTX 2080 Ti), dynamic RAG frameworks (+60% retrieval efficiency), YOLO aerial object detection (TF Lite GPU), and offline Android GGUF RAG chatbots.
Data Science & AI Engineering Analytics & Research Labs 2023 — 2024	Developed real-time financial price prediction models, digital receipt OCR extraction pipelines (+15% accuracy boost), and executive analytics dashboards.
Education & Foundation University CS Program	Bachelor of Science in Computer Science (B.S. CS) — Rigorous foundation in Software Engineering, Distributed Backend Systems, Machine Learning, and Computer Vision.